

AI-Based Alloy Manufacturing Analytics System

This project applies Artificial Intelligence and Machine Learning concepts to alloy ingot manufacturing. The system focuses on predicting alloy purity levels and waste material generation using industrial and IoT-based manufacturing data.

Business Problems

1. Predicting Alloy Purity Levels
2. Predicting Waste Material Generation

Project Objectives

- Improve alloy quality consistency
- Reduce rejected batches
- Minimise waste generation
- Improve manufacturing efficiency
- Support Industry 4.0 initiatives

Machine Learning Problem Types

- purity_pass → Classification
- purity_percentage → Regression
- waste_material_kg → Regression

Advanced Industrial Features

- Furnace temperature
- Furnace pressure
- Energy consumption
- Impurity levels
- Humidity level
- Machine vibration
- Slag percentage
- Maintenance frequency
- Downtime minutes
- Recycling rate
- IoT sensor monitoring

Business Value

- Improve product quality
- Reduce waste
- Lower operational costs
- Improve predictive maintenance
- Support smart manufacturing

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Jupyter Notebook
- GitHub

Future Improvements

- Real-time IoT integration
- Dashboard visualisation
- Cloud deployment
- Deep learning models

Target Variables Summary

Target Variable	Problem Type
purity_pass	Classification
purity_percentage	Regression
waste_material_kg	Regression

Conclusion

This project demonstrates how AI and machine learning can improve industrial manufacturing processes through predictive analytics, quality control, waste reduction, and operational intelligence.